



Boundary-spanning search and sustainable competitive advantage: The mediating roles of exploratory and exploitative innovations

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ABSTRACT

Although the critical role of boundary-spanning search in facilitating sustainable competitive advantage (SCA) has been recognized, how boundary-spanning search affects SCA under the conditions that competitors also search remains unclear. By considering competition, this study divides boundary-spanning search into proactive and responsive searches, and develops a theoretical model, in which proactive and responsive searches respectively influence exploratory and exploitative innovations, and subsequently enhance SCA. Empirical results from Chinese advanced manufacturing firms show that proactive and responsive searches have inverted U-shaped effects on SCA. Additionally, when the levels of proactive and responsive searches are low and moderate, exploratory innovation plays a complementary mediating role between proactive search and SCA, whereas exploitative innovation plays a complementary mediating role between responsive search and SCA. When considering competition, this study offers a crucial condition under which exploratory and exploitative innovations can transform the efforts of boundary-spanning search to SCA.

1. Introduction

It is well known that sustainable competitive advantage (SCA) plays a critical role in helping firms to gain excellent performance. According to resource-based view, resources are the primary sources of SCA. Hence, in order to achieve success, firms should acquire resources, especially kinds of knowledge resources (Mahdi, Nassar, & Almsafir, 2019). Boundary-spanning search, as an important way to acquire external knowledge, refers to the process that firms search new knowledge across organizational and technological boundaries. It can bring in novel and heterogeneous knowledge to accelerate technological advancements and prevent competitors' imitation. Therefore, boundary-spanning search has become a pivotal driver of firms' sustainability performance (Jung & Lee, 2016; Wang, Xue, & Yang, 2020).

As Greve and Taylor (2000) argued, firms search knowledge in the environment where competitors also search. In the context of dynamic competition, firms generally take competitors' actions into account when conducting search activities (Tang, Fu, & Yang, 2019). Accordingly, competitors' searches may influence firms' search strategies and outcomes. When considering competition, boundary-spanning search can be divided into proactive search and responsive search (Katila & Chen, 2008; O'Cass, Heirati, & Ngo, 2014). Specifically, proactive

search is the process that firms search external knowledge ahead of competitors to proactively cope with the changing environment; responsive search describes the process that firms search external knowledge by following competitors to responsively adapt to the changing environment.

Although the influence of competitors' searches on firms' searches has been recognized, how boundary-spanning search affects SCA under the conditions that competitors also search remains poorly understood. On the one hand, the main ideas of boundary-spanning search were developed in terms of firms' own experiences and past choices, such as search depth and search breadth (Laursen & Salter, 2006; Wang et al., 2020), and neglected the fact that competitors also search. The few studies that incorporate competition have inconsistent arguments on the impacts of proactive and responsive searches. For instance, Katila and Chen (2008) argued that proactive and responsive searches are positively related to firms' competitive advantage, though in different ways. However, other research argued that the returns of proactive and responsive searches are diminishing. As Baum, Li, and Usher (2000) suggested, proactive search is costly and time consuming, as early movers' searches are generally associated with risks and uncertainties. Meanwhile, Lieberman and Asaba (2006) demonstrated that responsive search raises the odds of core rigidity and cognitive myopia.

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On the other hand, while taking competitors' searches into consideration is important, it is also critical to orchestrate and leverage the knowledge obtained through boundary-spanning search to enhance SCA (Ngo, Bucic, Sinha, & Lu, 2019). Innovation is the process through which firms can deploy and utilize their knowledge to create or improve products and technologies (Aghion, Van Reenen, & Zingales, 2013). It can realize the potential value of firms' knowledge, and further facilitate firms' survival and competitive advantage (Schamberger, Cleven, & Brettel, 2013; Sharma, Davcik, & Pillai, 2016). Existing studies have explained the role of innovation in firms' success, but how innovation mediates the relationship between boundary-spanning search and SCA remains unclear.

The purpose of this study is to examine the relationships between proactive and responsive searches and SCA, and test the mediating role of innovation. According to ambidextrous theory, innovation can be divided into exploratory innovation and exploitative innovation (Chang, Hughes, & Hotho, 2011; He & Wong, 2004). As Auh and Menguc (2005) suggested, firms that proactively cope with the changing environment have orientations towards exploratory innovation, and firms that responsively adapt to the changing environment have orientations towards exploitative innovation. Hence, we will uncover how exploratory and exploitative innovations respectively mediate the effects of proactive and responsive searches on SCA.

Our research has two contributions. First, we divide boundary-spanning search into proactive and responsive searches by considering competition, and empirically test that both proactive and responsive searches have inverted U-shaped effects on SCA. Hence, our study enriches the literature of boundary-spanning search and SCA by incorporating competition, and provides reasonable explanations for the inconsistent arguments between proactive and responsive searches and firms' success. Second, we examine the mediating roles of exploratory innovation between proactive search and SCA, and exploitative innovation between responsive search and SCA. Our study elucidates the underlying mechanisms of how proactive and responsive searches link to SCA, and advances the understanding of the relationship between boundary-spanning search and SCA by integrating resource-based view with ambidextrous theory.

The remainder of this paper is structured as follows. First, we develop our hypotheses based upon a literature review. Second, we describe the research design, including sampling and data collection, variables and measures and so on. Third, we report the results of analyses, including descriptive statistics, correlations and hypotheses testing. Finally, the conclusions, implications and limitations are discussed.

2. Theoretical background and hypotheses

2.1. Boundary-spanning search and SCA

Resource-based view demonstrates that the inimitability and non-substitutability ensure the sustainability of competitive advantage. Proactive search is the process that firms search external knowledge ahead of competitors. According to the literature on "first mover advantage", proactive search enables firms to preferentially access to the knowledge from the outside (Katila & Chen, 2008). By integrating the newly obtained knowledge with old one, firms are able to enrich their knowledge base on which technological evolution relies (Jung & Lee, 2016). Hence, firms can take the lead to achieve technological advancements. Meanwhile, searching ahead of competitors also reduces competitors' opportunities to find valuable knowledge. As Peteraf (1993) argued, preferred access to knowledge, technological leads, and constraints on competitors' choices are the driving forces of inimitable positions. In this sense, proactive search facilitates the establishment of firms' inimitable positions, thus enhancing SCA.

However, some studies suggested that when hindering competitors' imitation, proactive search may induce the substitution threats from

competitors (Polidoro & Toh, 2011). The knowledge acquired through proactive search is generated in other firms or technological domains, and has the characteristics of novelty and originality (Asimakopoulou, Revilla, & Slavova, 2020). Polidoro and Toh (2011) stated that using novel knowledge in R&D activities gives rise to the substitution threats from competitors. This is because novel knowledge has some uncertainties in form and application, which increase competitors' difficulties to evaluate the validity and potential value of such knowledge. Consequently, competitors are encouraged to search alternative knowledge to produce substitutes. Therefore, in deterring imitation, proactive search also increases competitors' incentives to look for substitute knowledge, thus hurting SCA.

Given the irreconcilable relationship between inimitability and non-substitutability, we infer that the two forces jointly determine the relationship between proactive search and SCA. Firms conducting low levels of proactive search aim at external knowledge which is relatively close to existing knowledge, whereas firms conducting high levels of proactive search will target external knowledge which is distant from existing knowledge (Rosenkopf & Nerkar, 2001). As the level of proactive search increases, knowledge novelty and diversity increase, which strengthens firms' technological leads and inimitable positions. However, the increasing knowledge novelty also generates substitution threats from competitors. When the level of proactive search is too high, competitors will give up imitating and make efforts to create substitutes (Polidoro & Toh, 2011). Accordingly, before a certain point, the positive effects of inimitability are greater than the negative effects of substitutability on SCA, but after the point, the negative effects of substitutability will outweigh the positive effects of inimitability on SCA. Hence, this study proposes:

Hypothesis 1a. Proactive search has an inverted U-shaped effect on SCA.

Responsive search describes the process that firms search external knowledge by following competitors. It enables firms to gain legitimacy, and makes it easier to obtain knowledge beyond local (Jeong & Kim, 2019). Thus, responsive search accelerates the update of firms' knowledge pool, and allows firms to sustain existing competitive advantage while fostering new competitive advantage. As followers can learn from early movers' experiences, responsive search also improves firms' search efficiency (Lieberman & Asaba, 2006). Moreover, based on competitors' efforts, firms can gain a better understanding of the nature and application of new knowledge so as to improve products and processes (Asimakopoulou et al., 2020). Hence, firms will be more professional in current domains and more capable of maintaining competitive advantage.

Contrarily, responsive search also has some limitations. It intensifies competition in both knowledge and product markets (Posen, Yi, & Lee, 2020). On the one hand, searching external knowledge by following competitors drives up the price of knowledge. As the later comers, firms will encounter higher prices than early comers. On the other hand, responsive search leads to similar products between firms and their competitors, which increases firms' difficulties to build differentiated advantage (Asimakopoulou et al., 2020). Additionally, competitors' proactive exploration also restricts firms' search scope and opportunities to find valuable knowledge (Lieberman & Asaba, 2006). The insufficient input of external knowledge makes firms rely more heavily on existing knowledge, further leading to core rigidity and cognitive myopia.

As Deephouse (1999) argued, the relationship between strategic similarity and firms' performance depends on the relative strength of legitimacy and competition, and moderate levels of strategic similarity exert the greatest influence on firms' performance. By the similar logic, we infer that moderate levels of responsive search are optimal to SCA. As the level of responsive search increases from low to high, firms' legitimacy increases, which will generate some benefits to SCA, such as higher R&D efficiency. But there also comes intense competition, making firms hard to improve profitability and build differentiated

advantage. Chang and Park (2005) demonstrated that the legitimacy generated by following early movers increases at a decreasing rate, whereas competition increases at an increasing rate. Therefore, when the level of responsive search increases from low to moderate, SCA increases, because the positive effects of legitimacy are greater than the negative effects of competition on SCA. When the level of responsive search increases from moderate to high, the benefits of legitimacy start to be outweighed by the damages of competition. That is, the opposite effects initially lead to the increase of SCA, and then inflict the decrease of SCA. Thus, this study hypothesizes:

Hypothesis 1b. Responsive search has an inverted U-shaped effect on SCA.

2.2. Boundary-spanning search and exploratory and exploitative innovations

Exploratory innovation is a revolutionary activity that emphasizes the breakthrough of technological trajectories, the development of new products, and the expansion of original markets (He & Wong, 2004). Through proactive search, firms can obtain original and novel knowledge to create new ideas and expand technological change scopes (Katila & Chen, 2008). The combination of newly obtained and existing knowledge also accelerates the update of conventional routines (Jung & Lee, 2016). Meanwhile, firms that conduct proactive search have priority access to the information about consumers' potential needs and technological development trends (Lin & Li, 2013). This encourages firms to venture into new product and service development ahead of competitors and customers' cognition (Lisboa, Skarmetas, & Lages, 2011). Taken together, proactive search is paramount for exploratory innovation.

However, some research demonstrated that proactive search may harm the efficiency of exploratory innovation. The differences in terms of knowledge endowment, knowledge structure and knowledge cognition between firms give rise to knowledge distance, which makes the knowledge transfer and utilization in cross-boundary activities more difficult (Cummings & Teng, 2003). In this sense, firms that conduct proactive search will have difficulty in understanding the form and application of external knowledge (Asimakopoulou et al., 2020). As a result, there will be some uncertainties and risks in leveraging such knowledge. These uncertainties and risks not only reduce innovation validity, but also increase innovation costs, as firms have to pay more money, time, and efforts to absorb such knowledge (Linus, Siobhan, & Gann, 2016).

As the level of proactive search increases, the path-breaking novelty increases, but the innovation efficiency decreases. Specifically, when the level of proactive search increases from low to moderate, the creative thinking and path-breaking novelty increase, thus promoting exploratory innovation. Although there also come some risks and uncertainties reducing innovation efficiency, firms can mitigate these disadvantages through integration activities (Enkel, Groemminger, & Heil, 2018). Hence, from low to moderate level, the positive effects of proactive search on exploratory innovation are greater than its negative effects. However, as the level of proactive search increases from moderate to high, firms have to pay more efforts to reduce the negative influence of the decreasing efficiency, which not only distracts managers' attention, but also adds innovation costs. Accordingly, the benefits of proactive search will be offset and eventually exceeded by the costs of it. In sum, the likelihood that firms achieve exploratory innovation will initially increase, but then decrease with the increasing level of proactive search. Thus, this study expects:

Hypothesis 2a. Proactive search has an inverted U-shaped effect on exploratory innovation.

Exploitative innovation is an evolutionary activity that emphasizes the refinement of technologies, the adjustment of production processes,

and the reinforcement of existing market position (Chang et al., 2011). By following competitors, firms can access to more refined and mature knowledge (Katila & Chen, 2008). Hence, responsive search helps to reduce risks and errors in external knowledge utilization, and speeds the improvement of product and technology. Since early movers' experiences are good paradigms about what and how to search, responsive search also enables firms to focus on valuable knowledge (Lieberman & Asaba, 2006). Furthermore, by modifying competitors' searches, firms can accurately capture market and technology trends (Lisboa et al., 2011). In this sense, responsive search has some advantages, based on which firms can strengthen their skills and become more professional in current domains.

Besides the above positive effects, responsive search also has some negative effects on exploitative innovation. First, following competitors to search has time lags, hence the searched knowledge may be outdated (Baum et al., 2000). Second, since firms differ in knowledge endowments and requirements (Mickiewicz, Nyakudya, Theodorakopoulos, & Hart, 2017), the knowledge which is useful for some firms may be useless for others. In this sense, responsive search may obtain wrong knowledge (Posen et al., 2020). Third, relying too much on competitors' searches distracts firms' efforts to learn from other innovation partners, such as suppliers and independent experts (Schamberger et al., 2013). Thus, it will be difficult for firms to comprehensively assess the direction of product and technology development, thus dampening exploitative innovation.

These inconsistent opinions suggest that the relationship between responsive search and exploitative innovation may be non-linear. Firms with low levels of responsive search aim at exploring external knowledge from a broader scope rather than only by following competitors, whereas firms with high levels of responsive search may take competitors as the primary benchmarks (Sorensen & Stuart, 2000). As the level of responsive search increases, risks and errors decrease. Hence, firms are able to identify and integrate refined and mature knowledge to achieve exploitative innovation (Lieberman & Asaba, 2006). However, with firms relying more heavily on competitors' searches, they will be less likely to comprehensively assess the fast-changing environment and to obtain useful knowledge (Posen et al., 2020). Combined with the disadvantages resulting from time lags, the negative effects of responsive search will dominant the relationship between responsive search and exploitative innovation. Taken together, the positive effects of responsive search on exploitative innovation will increase at first, but after a certain point, it will decrease and gradually disappear. Thus, this study assumes:

Hypothesis 2b. Responsive search has an inverted U-shaped effect on exploitative innovation.

2.3. The mediating roles of exploratory and exploitative innovations

According to resource-based view, resource search and innovation are complementary in enhancing SCA (Laursen & Salter, 2006; Mahdi et al., 2019). Therefore, proactive and responsive searches are necessary but not sufficient for SCA, and they should be effectively converted into SCA through innovation. Given that proactive search gives rise to more revolutionary changes whereas responsive search brings in more evolutionary changes (Auh & Menguc, 2005; He & Wong, 2004), we infer that exploratory innovation mediates the effect of proactive search on SCA, whereas exploitative innovation mediates the effect of responsive search on SCA.

As the level of proactive search increases, firms can take the lead to access to the novel knowledge and the information about customers' potential needs. These knowledge and information enable firms to produce new ideas and capture product development trends (Jung & Lee, 2016). Hence, firms increase the chance to achieve exploratory innovation. Meanwhile, by adopting path-breaking technologies and commercializing innovative products, firms can expand their production

lines to new market and build differentiated advantage (Asimakopoulos et al., 2020). In this sense, exploratory innovation facilitates the transformation of proactive search to SCA. However, if proactive search exceeds a certain level, firms will suffer the damages caused by risks and uncertainties. These damages not only add the costs in assimilating and leveraging external knowledge, but also reduce the validity and efficiency of exploratory innovation (Linus et al., 2016). Consequently, it will be hard for firms to keep their technological leadership and superior profitability, eventually hurting sustainability performance. Thus, this study proposes:

Hypothesis 3a. Exploratory innovation mediates the inverted U-shaped effect of proactive search on SCA.

With the increase of the level of responsive search, firms can obtain more refined and mature knowledge (Katila & Chen, 2008). Hence, responsive search helps firms to update technologies and strengthen existing skills, and further promotes exploitative innovation. Meanwhile, exploitative innovation enables firms to be more professional in current fields and to better meet customers' needs (Chang et al., 2011). This not only underpins firms' existing competitive advantage, but also supports the development of new competitive advantage. Therefore, exploitative innovation accelerates the transformation of responsive search to SCA. However, when the level of responsive search is too high, firms may search in unsuitable ways and ultimately obtain useless knowledge (Lieberman & Asaba, 2006). Moreover, time lags resulting from high-level responsive search will weaken firms' capabilities to meet customers' new needs. Consequently, firms' products are no longer popular in current market, and firms would gradually lose their existing market share, which erodes their sustainability performance. Hence, this study hypothesizes:

Hypothesis 3b. Exploitative innovation mediates the inverted U-shaped effect of responsive search on SCA.

Fig. 1 shows the theoretical model of this study.

3. Research design

3.1. Sampling and data collection

This study collected data from Chinese advanced manufacturing firms, which include energy-saving and new energy vehicles, aerospace equipment, marine engineering equipment, high-speed railway transportation and so on. These advanced manufacturing firms were selected for two reasons. First, China has a big size of advanced manufacturing industry, and innovation in this industry is fast (Bao, Li, Pang, Bao, & Yi, 2017). The pressure resulting from rapid innovation encourages firms to pay more attention to enhancing innovation capability to maintain competitive advantage. Second, the Ministry of Industry and Information Technology of China has enacted "Made in China 2025" 1 + X Planning System. It regards the development of advanced

manufacturing as a key to strengthen the industrial capability of China, and provides a lot of policies supporting this industry. Therefore, firms in this industry have more opportunities to achieve sustainability performance.

We selected samples in Beijing, Shanghai, and four provinces in China, including Guangdong, Jiangsu, Shaanxi and Henan. Chinese Advanced Manufacturing City Development Index in 2019 reported that the advanced manufacturing firms can be divided into three levels according to the development degree. Firms at the first level are the leaders in this industry. They are the key innovators of advanced technologies, and mainly located in Beijing, Shanghai and Guangdong province. Firms at the second level have their own uniqueness in some fields, but they are relatively weak in cluster scale compared to the first-level firms. Such firms belong to the developing firms, and are mainly located in Jiangsu and Shaanxi province. Firms at the third level have some deficiencies in technological development and talent advantage compared to the first two levels, but they have cost advantage and policy supports. Firms in Henan province are the representatives of the third-level firms. Accordingly, firms in the six regions represent the different development levels of Chinese advanced manufacturing firms. To guarantee the typicality of samples, we randomly selected 100 firms in each region referring to the local business directory, a total of 600 firms.

To conduct the empirical research, we adapted the three-step procedure proposed by Gerbing and Anderson (1988). We first designed a questionnaire in English by referencing existing scales, and translated it into Chinese through double and reverse translation procedures (Werner & Campbell, 1970). Second, we performed a pretest with 50 MBA students who served in advanced manufacturing firms. According to the protest results, we interviewed three scholars and five managers from advanced manufacturing firms to revise the questionnaire. Third, we invited the selected firms to participate our research through telephone and E-mail, and 307 firms agree to participate. Since senior and mid-level managers generally have a broad overview of firms' knowledge and innovation strategies (Markovic & Bagherzadeh, 2018), we issued the formal questionnaires to them via E-mail. Every two weeks, we sent a reminder to those who have not responded. After two reminders, 276 responses were received. By excluding responses with too many errors and missing data, a total of 245 effective responses were received, and the effective response rate reached 40.83%. Table 1 presents the sample characteristics.

3.2. Non-response bias, later-response bias and common method variance

To test the non-response bias, we compared the differences between respondent and non-respondent groups in firm size ($p = 0.725$), firm age ($p = 0.398$) and R&D intensity ($p = 0.502$) through *t*-tests (Armstrong & Overton, 1977). We also compared early respondents (before the first reminder) and later respondents (after the second reminder) in terms of firm size ($p = 0.204$), firm age ($p = 0.350$) and R&D intensity ($p =$

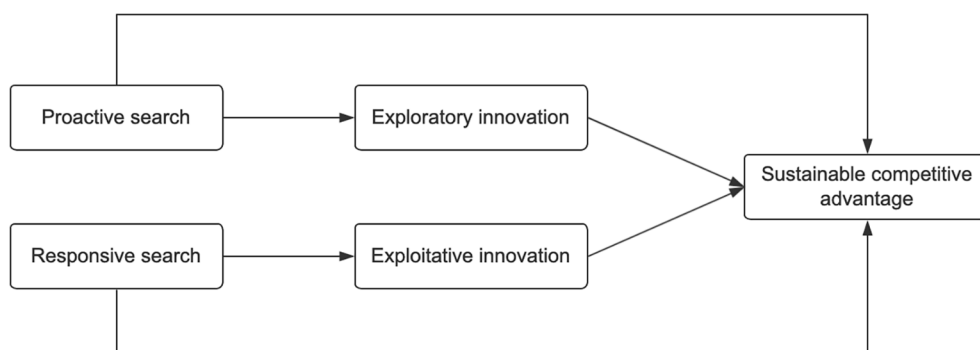


Fig. 1. Theoretical model.

Table 1
Sample characteristics (N = 245).

Characteristics	Frequency	Percentage	Characteristics	Frequency	Percentage
Firm size (employees)			Firm age (years)		
<100	21	8.57%	<5	25	10.20%
101–300	47	19.18%	6–10	28	11.43%
301–1000	92	37.55%	11–15	62	25.31%
1001–2000	70	28.57%	16–20	78	31.84%
>2000	15	6.13%	>20	52	21.22%
R&D intensity (%)			Annual sales (million RMB)		
<5	13	5.31%	<5	12	4.89%
5–10	41	16.73%	5–10	22	8.98%
11–15	99	40.41%	11–20	30	12.24%
16–20	88	35.92%	21–50	36	14.69%
>20	4	1.63%	51–100	47	19.18%
Firm ownership			101–200	56	22.87%
State-owned	38	15.51%	201–500	29	11.84%
Private-owned	123	50.21%	>500	13	5.31%
Mix-owned	72	29.39%			
Foreign-owned	12	4.89%			

0.645). The results indicated that neither non-response bias nor later-response is not a serious concern.

To examine the common method variance (CMV) deriving from single respondent, we compared three nested models (i.e., one-factor model, five-factor model, and five-factor with a common method factor model) through confirmatory factor analysis (CFA) (Podsakoff, MacKenzie, & Podsakoff, 2012). For one-factor model, all items of constructs were assigned to one factor. Its fit indices were not up to standard ($\chi^2 = 867.675$, $df = 209$, $RMR = 0.097$, $GFI = 0.725$, $TLI = 0.682$, $CFI = 0.713$, $RMSEA = 0.114$). For five-factor model, items were loaded on their separate theoretical constructs. This model fitted the data well ($\chi^2 = 306.762$, $df = 199$, $RMR = 0.056$, $GFI = 0.901$, $TLI = 0.945$, $CFI = 0.953$, $RMSEA = 0.047$). Based on five-factor model, we added a latent common method factor. The fit indices of this model were slightly improved ($\Delta RMR = 0.01$, $\Delta GFI = 0.014$, $\Delta TLI = 0.009$, $\Delta CFI = 0.012$, $\Delta RMSEA = 0.004$). But the common method factor accounted for only 3.57% of the total variance, and the factor loadings were still significant. That is, measurement models with or without the latent factor suggested no significant differences. Hence, CMV is not a problem in this study.

3.3. Variables and measures

This study adapted existing scales, and then invited three scholars and five managers in advanced manufacturing firms to revise these scales according to our study's context (see Table 2). Each item was measured on a five-point Likert scale. Drawn on the research of Katila and Chen (2008) and O'Cass et al. (2014), four items to measure proactive search and four items to measure responsive search were designed. Exploratory innovation was measured by the four-item scale adapted from He and Wong (2004), and exploitative innovation was measured by the four-item scale adapted from Chang et al. (2011). SCA refers to the competitive advantage that competitors cannot duplicate. To measure SCA, we used the six-item scale adapted from Wiggins and Ruefli (2002) and Chang (2011).

Existing research argued that several control variables may affect SCA. First, we considered firm size as a control variable (Wang et al., 2020), and adopted a generally accepted measurement—number of employees. Second, since firm age has influence on organizational behavior and advantage growth (Mikalef & Krogstie, 2020), we controlled for firm age and measured it by the years that a firm has been established. Third, given that R&D intensity has critical influence on innovation (Moaniba, Su, & Lee, 2020), we controlled R&D intensity and measured it by the ratio of R&D investments to sales. Fourth, as sales can provide a lot of supports for firms to perform innovative activities (Li, Li, Yu, & Yuan, 2019), we controlled for firm sales and measured it

Table 2
Reliability and convergent validity.

Items	Loadings
Proactive search (PS) (Katila & Chen, 2008; O'Cass et al., 2014) ($\alpha = 0.807$, AVE = 0.514, CR = 0.808)	
1. My firm discovers opportunities in new market or targets new customers ahead of competitors	0.660
2. My firm invents new products with unique features ahead of competitors	0.775
3. My firm discovers new ways to meet customer's needs ahead of competitors	0.674
4. My firm acquires entirely new product development skills and processes ahead of competitors	0.752
Responsive search (RS) (Katila & Chen, 2008; O'Cass et al., 2014) ($\alpha = 0.802$, AVE = 0.507, CR = 0.803)	
1. My firm strengthens its existing position in current markets by following competitors	0.803
2. My firm improves the efficiency of current products by following competitors	0.657
3. My firm improves the quality of current products by following competitors	0.700
4. My firm improves its product development processes and skills by following competitors	0.678
Exploratory innovation (EOR) (He & Wong, 2004) ($\alpha = 0.817$, AVE = 0.537, CR = 0.821)	
1. My firm focuses on introducing new generation of products	0.790
2. My firm focuses on extending product ranges	0.723
3. My firm focuses on opening up new markets	0.792
4. My firm focuses on entering new technology fields	0.611
Exploitative innovation (EOI) (Chang et al., 2011) ($\alpha = 0.811$, AVE = 0.520, CR = 0.812)	
1. My firm focuses on improving the provision efficiency of products	0.714
2. My firm focuses on increasing economies of scales in existing market	0.723
3. My firm focuses on expanding services for existing customers	0.682
4. My firm focuses on lowering costs of internal processes	0.763
Sustainable competitive advantage (SCA) (Chang, 2011; Wiggins & Ruefli, 2002) ($\alpha = 0.861$, AVE = 0.518, CR = 0.864)	
1. The quality of my firm's products is better than that of competitors	0.732
2. My firm's R&D capability is higher than competitors'	0.713
3. My firm's managerial capability is better than competitors'	0.622
4. My firm's profitability is better than competitors'	0.873
5. My firm's corporate image is better than competitors'	0.613
6. It is difficult for competitors to replace my firm's competitive position	0.733

by annual sales. Finally, firms with different ownerships have different strategic orientations, which may lead to different business modes (Aghion et al., 2013). Our samples were pertained to four ownerships (i.e., State-owned, Private-owned, Mix-owned and Foreign-owned). Therefore, we controlled for firm ownership and coded three dummy variables in the regression analyses, with Foreign-owned as the reference group.

3.4. Reliability and validity

This study used Cronbach's α and composite reliability (CR) to test the reliability of construct variables (Hair, Ringle, & Sarstedt, 2011). As presented in Table 2, the values of Cronbach's α and CR of construct variables were greater than the minimum threshold 0.7, indicating a satisfactory reliability.

Convergent validity was assessed by item loadings and average variance extracted (AVE) (Fornell & Larcker, 1981). CFA with all items that pertain to their theoretical constructs yielded a model fitting the data reasonably ($\chi^2 = 306.762$, $df = 199$, $RMR = 0.056$, $GFI = 0.901$, $TLI = 0.945$, $CFI = 0.953$, $RMSEA = 0.047$), and item loadings were greater than 0.5 ($p < 0.001$). Moreover, the AVE values of construct variables were bigger than the suggesting value 0.5. Hence, results listed in Table 2 provided an acceptable convergent validity.

To test discriminant validity, we compared different nested models through CFA (Fornell & Larcker, 1981). As presented in Table 3, five-factor model was the best, and its fit indices were acceptable, inferring an adequate discriminant validity. Additionally, as shown in Table 4, the square roots of AVE were greater than the correlation coefficients among construct variables. Therefore, the discriminant validity was supported.

4. Results

4.1. Descriptive statistics and correlations

Table 4 summarizes descriptive statistics and correlations of all variables in this study. The results indicate significant positive correlations among construct variables.

4.2. Hypotheses testing

4.2.1. Direct effects

Following the suggestion of Hanns, Pieters, and He (2016), the inverted U-shaped effect of independent variable X on dependent variable Y can be expressed as: $Y = \beta_0 + \beta_1 X + \beta_2 X^2$, and the establishment of an inverted U-shaped relationship can be tested through a three-step procedure. First, a significant and negative β_2 should exist. Second, the slope at both ends of the X-range should be sufficiently steep. Specifically, the slope at X_L , the low end of the X-range, which is $\beta_1 + 2\beta_2 X_L$, should be positive and significant. And the slope at X_H , the high end of the X-range, which is $\beta_1 + 2\beta_2 X_H$, should be negative and significant. Third, the turning point $-\beta_1/2\beta_2$ should be located within the X-range.

Hypotheses 1a and 1b posit that proactive and responsive searches have inverted U-shaped effects on SCA. Model 3 in Table 5 shows that proactive search squared ($\beta = -0.179$, $p < 0.01$) and responsive search squared ($\beta = -0.140$, $p < 0.05$) are negatively and significantly related to SCA. After standardized, proactive search ranges from -2.170 to

1.580 , and responsive ranges from -2.364 to 1.636 . As for proactive search, the slope at -2.170 , which is 1.119 , is positive and significant; the slope at 1.580 , which is -0.224 , is negative and significant. As for responsive search, the slope at -2.364 , which is 0.779 , is positive and significant; the slope at 1.636 , which is -0.341 , is negative and significant. Moreover, the turning point of the relationship between proactive search and SCA is 0.955 , and 0.418 for the relationship between responsive search and SCA. Both are located within the range of proactive and responsive searches, respectively. Hence, hypotheses 1a and 1b are supported.

Hypothesis 2a posits that proactive search has an inverted U-shaped effect on exploratory innovation. Model 7 in Table 5 shows that proactive search squared ($\beta = -0.306$, $p < 0.001$) is negatively and significantly related to exploratory innovation. The slope at -2.170 , which is 1.585 , is positive and significant; the slope at 1.580 , which is -0.710 , is negative and significant. The turning point is 0.420 , which is located within the range of proactive search. Hence, hypothesis 2a is supported.

Hypothesis 2b posits that responsive search has an inverted U-shaped effect on exploitative innovation. Model 10 in Table 5 reports that responsive search squared ($\beta = -0.136$, $p < 0.05$) is negatively and significantly related to exploitative innovation. The slope at -2.364 , which is 0.894 , is positive and significant; the slope at 1.636 , which is -0.194 , is negative and significant. The turning point is 0.923 , which is located within the range of responsive search. Hence, hypothesis 2b is supported.

4.2.2. Mediating effects

The causal steps approach for testing mediating effects may distort the relationships between variables, as it assumes that relationships linking X, mediating variable M, and Y are linear (Baron & Kenny, 1986). Therefore, this study adopted the method of Hayes and Preacher (2010) to estimate the mediating effects in models which are nonlinear. As Hayes and Preacher described, the rate at which a change in X changes Y indirectly through the changes in M, expressed here as θ , is called the instantaneous indirect effect of X on Y through M. It can be estimated as the product of the first partial derivative of the functions of M on X and Y on M. In this study, the models of M and Y can be denoted as Eqs. (1) and (2). Consequently, we obtained the instantaneous indirect effect of X on Y through M as Eq. (3).

$$M = i_1 + a_1 X + a_2 X^2 \quad (1)$$

$$Y = i_2 + c'_1 X + c'_2 X^2 + bM \quad (2)$$

$$\theta = \frac{\partial M}{\partial X} \cdot \frac{\partial Y}{\partial M} = (a_1 + 2a_2 X)b \quad (3)$$

Although structural equation model (SEM) can include different constructs in one model, it estimates the indirect and direct effects simultaneously instead of assuming that Eqs. (1) and (2) are independent (Iacobucci, 2008). That is, SEM may be unsuitable to estimate instantaneous indirect effect. Hence, we performed regression analyses to test the mediating effects. By conducting MEDCURVE macro for SPSS, the instantaneous indirect effects at the mean of X values as well as plus and minus a standard deviation from the mean, which represent low, moderate and high levels of X, were yielded. In addition, through the resample analysis of 1000 bootstrap samples, the 95% bias-corrected bootstrap confidence intervals (CIs) of the instantaneous indirect effects were illustrated.

Hypothesis 3a posits that exploratory innovation mediates the inverted U-shaped effect of proactive search on SCA. Table 6 reports that the values of the instantaneous indirect effect for low, moderate, and high levels of proactive search are 0.252 , 0.097 and -0.058 . At low and moderate levels of proactive search, proactive search can influence SCA via exploratory innovation, as the 95% bootstrap CIs of instantaneous indirect effects are above 0, although θ reduced from 0.252 to 0.097 . However, at high levels of proactive search, proactive search has no

Table 3
Discriminant validity and CMV test.

Models	χ^2	df	RMR	GFI	TLI	CFI	RMSEA
One-factor model ^a	867.675	209	0.097	0.725	0.682	0.713	0.114
Three-factor model ^b	586.240	206	0.075	0.796	0.814	0.834	0.087
Four-factor model ^c	458.749	203	0.070	0.843	0.873	0.888	0.072
Four-factor model ^d	434.267	203	0.062	0.848	0.885	0.899	0.068
Five-factor model	306.762	199	0.056	0.901	0.945	0.953	0.047
Five-factor factor + CMV	257.771	177	0.046	0.915	0.954	0.965	0.043

Notes: ^a PS + RS + EOR + EOI + SCA; ^b PS + RS, EOR + EOI, SCA; ^c PS + RS, EOR, EOI, SCA; ^d PS, RS, EOR + EOI, SCA.

Table 4
Descriptive statistics and correlations.

Variables	1	2	3	4	5	6	7	8	9	10	11	12
1. Firm size	—											
2. Firm age	0.297***	—										
3. R&D intensity	0.352***	0.199***	—									
4. Annual sales	−0.072	0.028	0.039	—								
5. Ownership 1	0.014	0.054	−0.014	−0.038	—							
6. Ownership 2	0.012	0.058	−0.111	−0.011	−0.430***	—						
7. Ownership 3	0.015	−0.113	0.114	0.072	−0.276***	0.648***	—					
8. PS	0.099	−0.001	0.266***	0.052	0.033	−0.166**	0.164*	0.717				
9. RS	0.087	−0.023	0.144*	0.088	−0.055	−0.080	0.096	0.439***	0.712			
10. EOR	0.252***	0.055	0.263***	0.036	0.074	−0.138*	0.070	0.450***	0.360***	0.733		
11. EOI	0.214***	0.131*	0.182**	0.033	0.007	−0.071	0.069	0.424***	0.340***	0.508***	0.721	
12. SCA	0.216***	0.067	0.361***	0.103	0.012	−0.118	0.067	0.582***	0.438***	0.576***	0.528***	0.720
Mean	3.045	3.424	3.265	4.767	0.155	0.502	0.294	3.420	3.364	3.482	3.459	3.525
S.D.	1.033	1.231	0.858	1.835	0.363	0.501	0.456	0.873	0.796	0.872	0.832	0.829

Notes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; numbers in bold on the diagonal indicate the square root of AVE.

Table 5
Regression results.

Variables	SCA				EOR			EOI		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9	Model 10
Control variables										
Firm size	0.137*	0.110*	0.099	0.013	0.208**	0.197**	0.198**	0.162*	0.137*	0.125
Firm age	−0.036	−0.003	−0.033	−0.042	−0.043	−0.024	−0.065	0.068	0.088	0.084
R&D intensity	0.306***	0.174**	0.171**	0.209***	0.185**	0.085	0.065	0.101	0.065	0.069
Annual sales	0.108	0.073	0.081	0.080	0.048	0.033	0.031	0.036	0.008	0.015
Ownership 1	−0.171	−0.166	−0.156	−0.152	−0.033	−0.061	−0.048	−0.022	0.032	0.037
Ownership 2	−0.300*	−0.249*	−0.232*	−0.210	−0.194	−0.187	−0.182	−0.069	−0.001	0.017
Ownership 3	−0.224	−0.259*	−0.241*	−0.192	−0.098	−0.152	−0.145	0.010	0.048	0.062
Independent variables										
PS		0.440***	0.342***			0.402***	0.257***			
PS ²			−0.179**				−0.306***			
RS		0.200***	0.117*						0.317***	0.251***
RS ²			−0.140*							−0.136*
Mediators										
EOR				0.356***						
EOI				0.309***						
R ²	0.167	0.448	0.495	0.461	0.118	0.264	0.332	0.069	0.164	0.147
R ² change	—	0.281	0.047	0.294	—	0.146	0.068	—	0.095	0.013
F	6.787***	21.155***	20.768***	22.365**	4.529***	10.577***	12.992***	2.492*	5.797***	5.654***

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 6
Instantaneous indirect effects.

Mediators	Independent variable values	95% confidence intervals		Instantaneous indirect effects
		Lower CI	Upper CI	
Exploratory innovation	Proactive search	2.547	0.149	0.360
		3.420	0.046	0.169
		4.293	−0.159	0.026
Exploitative innovation	Responsive search	2.568	0.104	0.324
		3.364	0.054	0.193
		4.160	−0.068	0.143

influence on SCA through its influence on exploratory innovation, as 0 is inside the 95% bootstrap CIs. In sum, as the level of proactive search increases, the mediating effect of exploratory innovation decreases and ultimately becomes insignificant. Hence, hypothesis 3a is partially supported.

Hypothesis 3b posits that exploitative innovation mediates the inverted U-shaped effect of responsive search and SCA. As Table 6 presented, the values of the instantaneous indirect effect for low, moderate, and high levels of responsive search are 0.199, 0.111 and 0.022. At low and moderate levels of responsive search, responsive search can enhance SCA via exploitative innovation, as the 95% bootstrap CIs of

instantaneous indirect effects are above 0, although there is a diminishing return. However, at high levels of responsive search, responsive search has no influence on SCA through its influence on exploitative innovation, as 0 is inside the 95% bootstrap CIs. That is, as the level of responsive search increases, the mediating effect of exploitative innovation decreases and ultimately becomes insignificant. Hence, hypothesis 3b is partially supported.

Baron and Kenny (1986) classification of full-partial-no mediation assumes that the indirect and direct paths are one-dimensional. Zhao, Lynch, and Chen (2010) suggested that regarding indirect and direct paths as two dimensions can better interpret the conception of mediation. Following the ideas of Zhao et al. (2010), mediation can be classified into five types (i.e., complementary mediation, competitive mediation, indirective mediation, direct-only nonmediation, and no-effect nonmediation).

The results of the mediating effect test show that in the regression of SCA on proactive search, proactive search squared and exploratory innovation, the coefficients of proactive search ($\beta = 0.331$, $p < 0.001$), proactive search squared ($\beta = -0.129$, $p < 0.01$) and exploratory innovation ($\beta = 0.329$, $p < 0.001$) are significant. And in the regression of SCA on responsive search, responsive search squared and exploitative innovation, the coefficients of responsive search ($\beta = 0.211$, $p < 0.001$), responsive search squared ($\beta = -0.184$, $p < 0.001$) and exploitative innovation ($\beta = 0.399$, $p < 0.001$) are also significant. Hence, proactive

and responsive searches have direct effects on SCA. When the levels of proactive and responsive searches are high, there are no indirect effects between proactive and responsive searches and SCA. Thus, in this situation, the mediation types of exploratory and exploitative innovations are both direct-only nonmediation. When the levels of proactive search are low and moderate, proactive search has inverted U-shaped effects on exploratory innovation and SCA, and exploratory innovation has positive effects on SCA. The directions of indirect and direct effects are same. Thus, in this case, the mediation type of exploratory innovation is complementary mediation. When the levels of responsive search are low and moderate, responsive search has inverted U-shaped effects on exploitative innovation and SCA, and exploitative innovation has positive effects on SCA. The directions of indirect and direct effects are same. Therefore, in this case, the mediation type of exploitative innovation is complementary mediation.

5. Discussion

5.1. Conclusions

The purpose of this study is to examine how boundary-spanning search affects SCA under the conditions that competitors also search, and test the mediating roles of exploratory and exploitative innovations. Specifically, we divide boundary-spanning search into proactive search and responsive search by incorporating competition. The empirical results suggest that both proactive and responsive searches have inverted U-shaped effects on SCA. Moreover, proactive and responsive searches respectively exert inverted U-shaped effects on exploratory and exploitative innovations. Our results partially support the mediating roles of exploratory and exploitative innovations. Interestingly, when the levels of proactive and responsive searches are low and moderate, exploratory innovation plays a complementary mediating role between proactive search and SCA, whereas exploitative innovation plays a complementary mediating role between responsive search and SCA.

However, when the levels of proactive and responsive searches are high, the mediating roles of exploratory and exploitative innovations are not supported. The possible explanations may be that at high levels of proactive and responsive searches, exploratory and exploitative innovations are not the effective approaches to realize the potential value of external knowledge. On the one hand, when the level of proactive search is high, firms are capable of building knowledge barriers against competitors' imitation (Jung & Lee, 2016; Lieberman & Asaba, 2006). Based on this knowledge advantage, firms have more approach options to transform the obtained knowledge to SCA. To maximize returns, firms may select the approaches whose costs are lower than exploratory innovation. On the other hand, when the level of responsive search is high, firms may take their competitors as the only benchmarks, which limits firms' opportunities to develop innovation. Given that it is easier to modify competitors' innovation than to develop an original one (Polidoro & Toh, 2011), firms may leverage the obtained knowledge to modify competitors' innovation rather than to conduct exploitative innovation to sustain competitive advantage.

5.2. Theoretical contributions

Our findings contribute to the literature in three ways. First, we highlight the impact of competitors' searches on firms' search outcomes. Extant research mainly focused on the effect of boundary-spanning search on firms' performance from the perspective of firms' own experiences and past choices (Jung & Lee, 2016; Lin & Li, 2013), and neglected the fact that competitors also search. The few studies that incorporate competition have inconsistent arguments about the effects of boundary-spanning search (Baum et al., 2000; Katila & Chen, 2008; Lieberman & Asaba, 2006). By incorporating competition, we divide boundary-spanning search into proactive and responsive searches, and confirm that proactive and responsive searches have inverted U-shaped

effects on SCA. On the one hand, our study enriches the literature of boundary-spanning search by taking competitors' searches into account, and advances the understanding of the relationship between boundary-spanning search and SCA by incorporating competition. On the other hand, our study demonstrates the inverted U-shaped effects of proactive and responsive searches on SCA, and offers reasonable explanations to the inconsistency of prior work that some claimed the impact of boundary-spanning search on firms' success is positive, while others claimed it is negative.

Second, we theorize and examine the inverted U-shaped effects of proactive search on exploratory innovation, and responsive search on exploitative innovation. As Auh and Menguc (2005) suggested, in competitive environments, firms with different search strategies have different innovation orientations. However, prior studies that include competition only revealed the influence of proactive and responsive searches on product innovation, without distinguishing the innovation differences (Katila & Chen, 2008). To further explain the effects of boundary-spanning search on innovation, we elaborate the innovation differences resulting from proactive and responsive searches. Our study provides effective ways to convert proactive and responsive searches to SCA, and speaks to future research on the match of search strategies and innovation modes. Our study also enriches the research of the relationships between boundary-spanning search and exploratory and exploitative innovations by incorporating competition. Moreover, our results are partially in line with the research of Laursen and Salter (2006) and Wang et al. (2020), and support their arguments that the relationship between boundary-spanning search and innovation is curvilinear rather than linear.

Finally, we prove that when the levels of proactive and responsive searches are low and moderate, exploratory innovation plays a complementary mediating role between proactive search and SCA, whereas exploitative innovation plays a complementary mediating role between responsive search and SCA. Drawing on resource-based view and ambidextrous theory, we provide a new theoretical mechanism through which boundary-spanning search links to SCA. Our findings partially respond to Ngo et al. (2019), who treated exploratory and exploitative innovations as the outcomes of knowledge and the prerequisites of firms' success. However, when the levels of proactive and responsive searches are relatively high, the mediating roles of exploratory and exploitative innovations are not supported. As such, we offer a crucial condition under which exploratory and exploitative innovations can facilitate the transformation of boundary-spanning search to SCA. Meanwhile, our study explains an important research issue that when and how firms can benefit from boundary-spanning search and exploratory and exploitative innovations.

5.3. Managerial contributions

Our study makes three managerial contributions. First, when taking competitors' searches into consideration, both proactive and responsive searches have inverted U-shaped effects on SCA. Hence, to achieve sustainability performance, managers need to be aware of the influence of competitors' searches on firms' search outcomes. It is necessary to take competitors' searches into account when designing search strategies. Moreover, since knowledge from other firms and technological fields are truly critical for firms to establish technological lead positions and increase market share, managers should build effective communication and learning mechanisms with other firms and experts to access to new knowledge and technologies. Building strategic alliances and collaboration networks may help to improve the efficiency of external knowledge acquisition. However, given that high levels of proactive and responsive searches will exert negative influence on SCA, managers should control their search activities at an appropriate level. In addition to considering knowledge novelty, managers also should consider the search efficiency and knowledge reliability.

Second, our findings demonstrate the inverted U-shaped effects of

proactive search on exploratory innovation, and responsive search on exploitative search. Therefore, managers need to balance the positive and negative influence of proactive and responsive searches to achieve better innovation performance. They can build a knowledge evaluation mechanism to assess the potential value and integration costs of external knowledge and opportunities, so as to keep search activities at an appropriate level and maximize the benefits. Furthermore, our study also suggests that proactive and responsive searches can bring in different types of knowledge, which in turn promote different types of innovation. Hence, managers should match search activities with suitable innovation modes to effectively realize the potential value of external knowledge.

Finally, our study highlights that exploratory innovation mediates the effect of proactive search on SCA when the level of proactive search is low and moderate, and exploitative innovation mediates the effect of responsive search on SCA when the level of responsive search is low and moderate. Hence, managers should take steps to convert the efforts of search activities to SCA, and do well on the match of search strategies and innovation modes. However, when the levels of proactive and responsive searches are relatively high, the mediating roles of exploratory and exploitative innovations are not supported. In this situation, transforming the obtained knowledge to SCA through exploratory and exploitative innovations not only decreases transformation efficiency, but also adds knowledge utilization costs. Therefore, managers should select other approaches to realize the potential value of the obtained knowledge rather than exploratory and exploitative innovations.

5.4. Limitations and future research

Our study also has several limitations. First, the data used in empirical analyses were from Chinese advanced manufacturing firms, so the results may be limitable in other industries or countries. Therefore, future research could put these results in other industries and countries to examine the applicability of them. Moreover, the data were mainly collected through questionnaire survey. Different methodologies, such as focus-group interviews and secondary data, may help to improve research validity in future research.

Second, when the levels of proactive and responsive searches are relatively high, the mediating roles of exploratory and exploitative innovations are not supported. This implies that some mediators may be omitted in our research (Zhao et al., 2010). To fully explain the underlying mechanism of how boundary-spanning search affects SCA, future research can further investigate the omitted mediators.

Third, the importance of digital technologies, such as big data analytics, has been increasingly recognized given their pivotal roles in sensing emerging opportunities and increasing firms' reach of knowledge control (Erevelles, Fukawa, & Swayne, 2016; Mikalef & Krogstie, 2020). Hence, it will be of great significance to explore how digital technologies regulate the relationship between boundary-spanning search and firms' success. Future research can further investigate this interesting issue.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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